

SCHOOL ID:

NAME:

Midterm Examination Problem/Answer Sheet

TIME: 04/08/2025, 13:20–16:10

This is an open-book exam. You can use any printed materials as your reference during the exam. Any electronic devices are not allowed. Both English and Chinese (if suited) are allowed for answering the questions. We do not accept any other languages.

Any form of cheating, lying or plagiarism will not be tolerated. Students can get zero scores and/or get negative scores and/or fail the class and/or be kicked out of school and/or receive other punishments for those kinds of misconducts.

There are 10 problems. 5 of them are marked with * and are supposedly simple(r); 5 of them are marked with ** and are supposedly hard(er). Each problem is worth 10 points. **We will calculate your midterm exam score by re-weighting your best two problems by 150%—scaling them to 15 points each, and your worst two problems by 50%—scaling them to 5 points each.** For each problem, you can answer it with one and only one side, as the problem/answer sheet indicates. For the odd-numbered problems, please write down your SCHOOL ID (top-left) and your NAME (top-right) on sheet. Then, write down the SOLUTION to each problem as instructed. You should keep your answers as *concise* as possible to facilitate grading.

1. (*) Write down your SCHOOL ID and NAME above, and your SOLUTION below.

Happy April Fool's Day! The problems are still being cooked. Please stay tuned and good luck with all your midterms!

2. (**) Write down your SOLUTION below.

SCHOOL ID:

NAME:

3. (*) Write down your SCHOOL ID and NAME above, and your SOLUTION below.

4. (**) **Write down your SOLUTION below.**

SCHOOL ID:

NAME:

5. (*) Write down your SCHOOL ID and NAME above, and your SOLUTION below.

6. (**) **Write down your SOLUTION below.**

SCHOOL ID:

NAME:

7. (*) Write down your SCHOOL ID and NAME above, and your SOLUTION below.

8. (**) **Write down your SOLUTION below.**

SCHOOL ID:

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9. (*) Write down your SCHOOL ID and NAME above, and your SOLUTION below.

10. (**) **Write down your SOLUTION below.**